Integrating Cluster and Quantum Frameworks for High-Performance Computing

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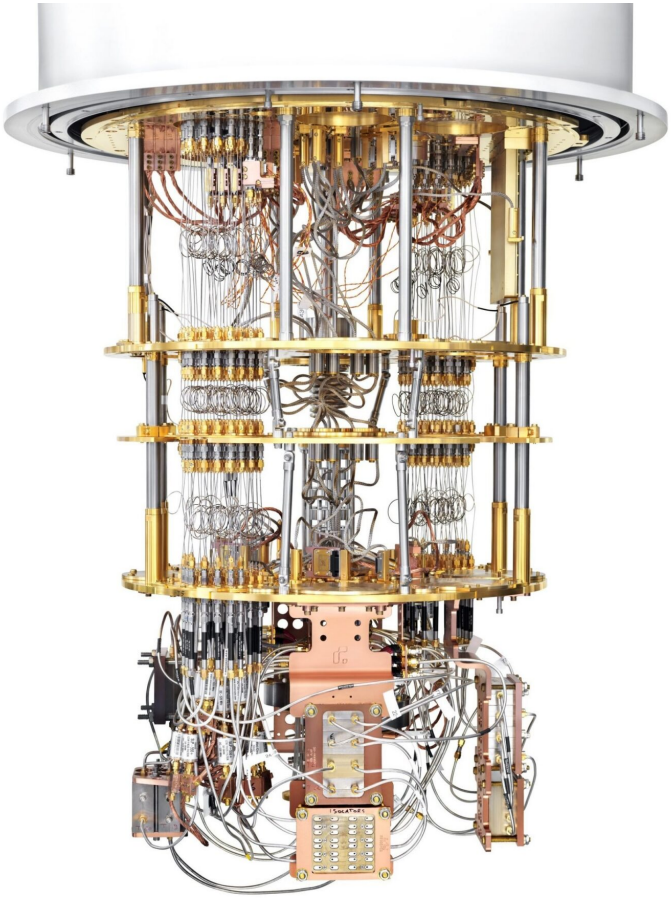
What is Quantum Computing?



Quantum Computers leverage principles of quantum mechanics to perform some computational tasks much more efficiently.

- Superposition
- Entanglement
- Interference
- Reversibility

Why Quantum Computing?

Source: Dr. David Root, *IEEE Microwave Theory and Technology Society*

Linear Algebra Refresher

Dirac Notation

- Ket: $|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$
- Bra: $\langle\psi| = \begin{pmatrix} \alpha^* & \beta^* \end{pmatrix}$
- Inner Product: $\langle\phi|\psi\rangle$

Tensor Products Used to represent multi-qubit systems:

$$|\psi\rangle \otimes |\phi\rangle = |\psi\phi\rangle$$

Conjugate Transpose

$$A^\dagger = (A^T)^*$$

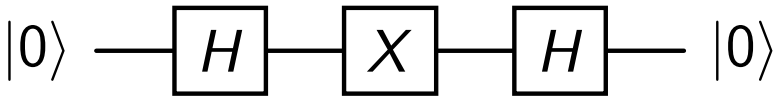
A gate U is valid if $U^\dagger U = I$.

Quantum Gates: The Ideal Physics World

High-Level Definition

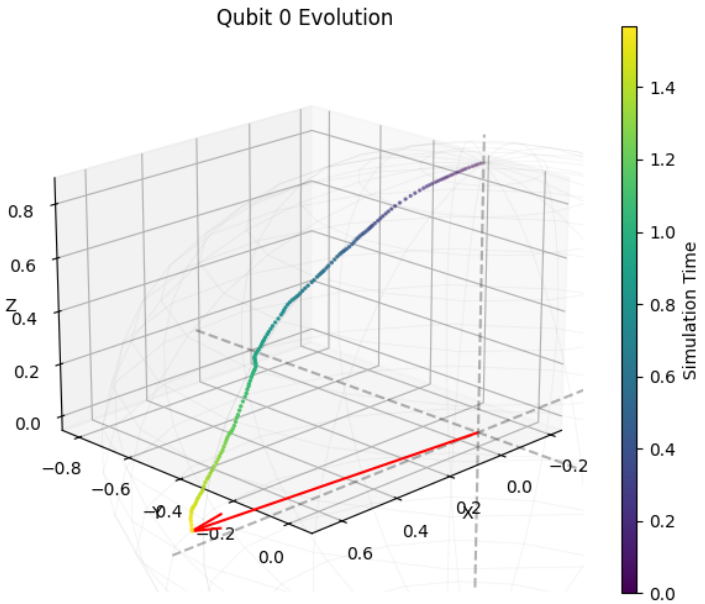
A quantum gate is a unitary transformation applied to a qubit state. In a "perfect" physics world, these operations are:

- **Deterministic:** Always performs the exact rotation intended.
- **Isolated:** No interaction with the outside environment.
- **Reversible:** No information is lost during computation.

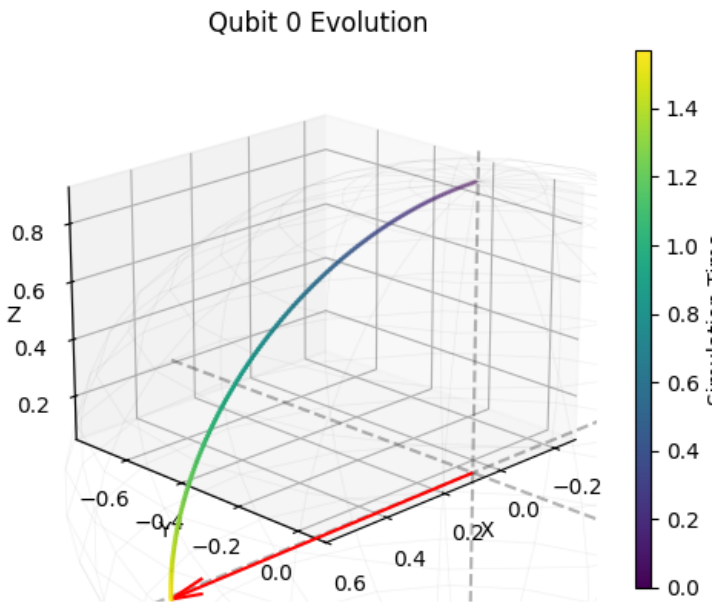


Comparing Ideal Gates to Noisy Reality

Noisy Output



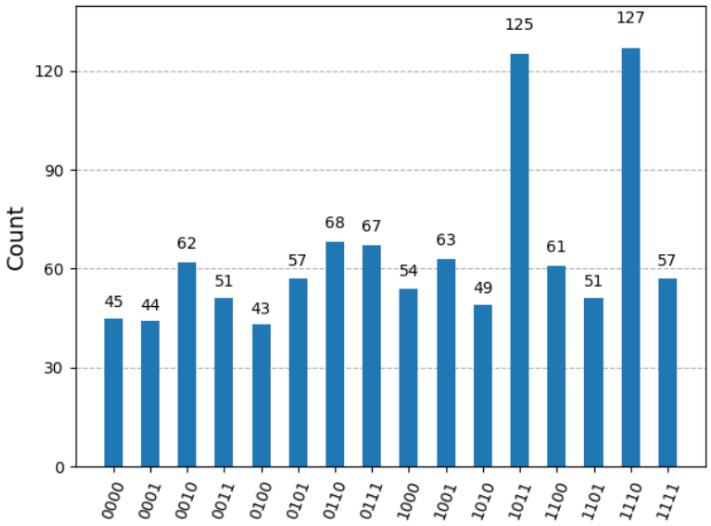
Ideal Output



- **Left:** Gate errors create a "wavey" appearance.
- **Right:** Theoretical simulation in a perfect environment.

Qiskit Simulation Architecture

- **Gate Subsections:** Modular circuit design for scalability.
- **Batching:** Running multiple "shots" to gather statistics on noisy outputs.



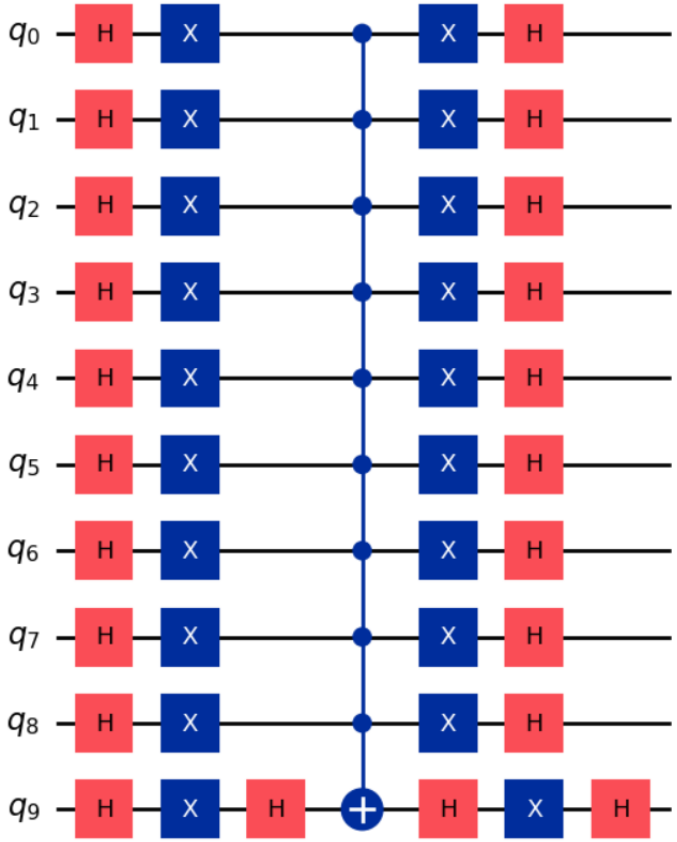


Figure: Qubit Diffuser Circuit

Command Line Interface (CLI)

In order to bridge the gap between the high level Qiskit simulations and low-level C phsysics simulation I developed a CLI to simulate perfect quantum gates.

Sample Usage

```
$ qit> create q1 10
$ qit> cnot q1 0 1 ## 0 is ctrl and 1 is targ
$ qit> entangle bp1 q1 2 3 ## bell pair
$ qit> measure q1 0 1 2 3
```

This also allows for easy integration with the C-based simulator, where we can directly tell the CLI to read from the QIT output.

Low-Level Simulation in C

Schrodinger Equation:

$$i\frac{d}{dt}|\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle$$

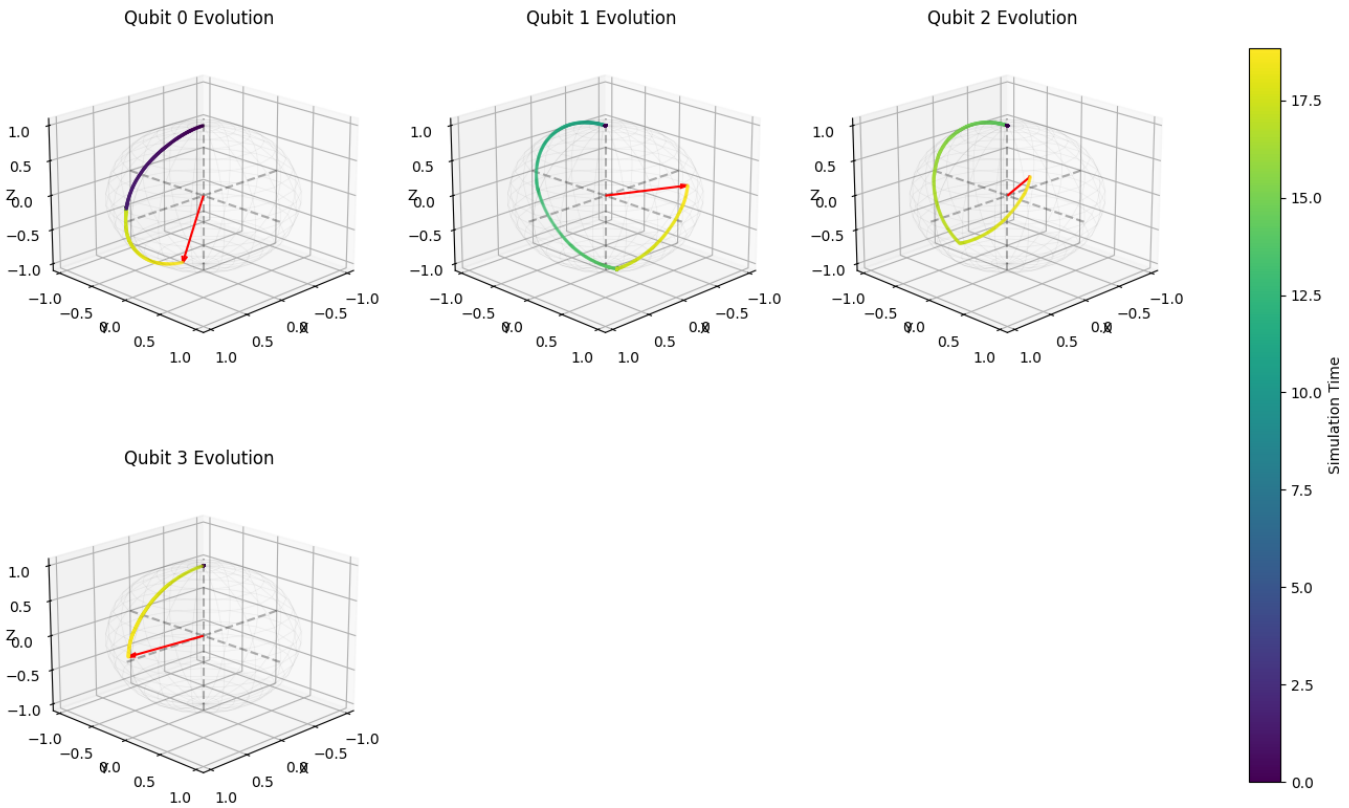
Single Qubit Hamiltonian:

$$\hat{H}_k = \frac{1}{2}\delta\hat{\sigma}_z + \frac{1}{2}\Omega(\cos(\phi)\hat{\sigma}_x + \sin(\phi)\hat{\sigma}_y)$$

Cross-Resonance Hamiltonian:

$$\hat{H}_{CR} = \frac{\Omega_{cr}}{2}(\hat{\sigma}_z^{(c)} \otimes \hat{\sigma}_x^{(t)})$$

Low-Level Simulation in C (Cont.)



Takeaways and Conclusions

- **Integration is Key:** Quantum computers won't replace clusters; they will act as specialized "accelerators."
- **Hardware vs. Simulation:** Simulation is essential for debugging QEC algorithms before running on expensive hardware.
  - **Example:** Simulating a noisy CNOT gate to understand error rates.
- **Future Work:** Optimizing the C-simulation with OpenMPI for massively parallel state-vector evolution.

Questions?